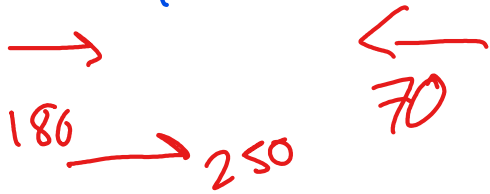


1000 km

$$180 + 70 \rightarrow 250 \text{ kmph}$$



bandwidth: 3Mbps

maximum amount of data that can be transferred over a connection per second

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$\text{Time} = \frac{D}{S} = \frac{1000 \text{ km}}{250} = 4 \text{ hours}$$

$$\begin{aligned} \text{Total Data} &= \text{Bandwidth} \times \text{Time} \\ &= 3 \times 4 \times 60 \times 60 \\ &= 5400 \text{ MBps} \\ &= 5.4 \text{ GB} \end{aligned}$$

$$1000 \text{ MB} = 1 \text{ GB}$$

ASCII 'a' = 97 decimal

$$\begin{array}{r} 2 \overline{) 97} \\ \underline{2 8} \\ 2 24 \\ \underline{2 12} \\ 2 6 \\ \underline{2 3} \\ 2 1 \\ \underline{2 0} \\ 0 \end{array}$$

binary 7 bits

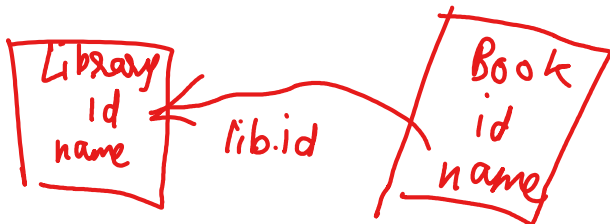
$$\begin{aligned} &1000011 \\ &2^0 + 2^5 + 2^6 \\ &1 + 32 + 64 = 97 \text{ decimal} \end{aligned}$$

1 revolution $\rightarrow$ 600 bits

MAX SPEED $\rightarrow$ 42,000 bits/second

RPM revolution per minute

$$\begin{array}{l} 42,000 \div 60 \rightarrow \text{read 1 revolution} = 700 \\ \text{RP Second} \end{array}$$



Bandwidth = $\frac{\text{Megabits (DATA)}}{\text{second}}$ $\Rightarrow$ hour

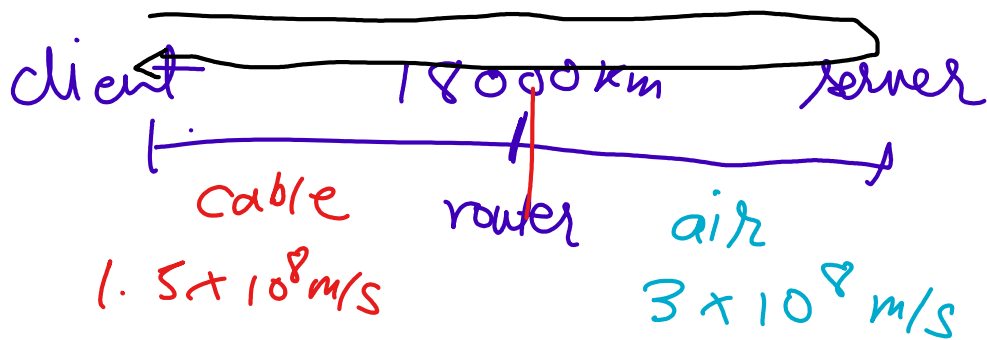
$$\text{User 1} = \frac{4 \times 8}{\text{hours bandwidth}} \rightarrow 32 \text{ Megabits}$$

$$\begin{aligned} \text{User 2} &= 4 \times 6 + 8 \times 2 \\ &= 24 + 16 = 40 \text{ Megabits} \end{aligned}$$

$$\begin{aligned} \text{User 3} &= 4 \times 2 + 4 \times 10 \\ &= 8 + 40 = 48 \text{ Megabits} \end{aligned}$$

$$\begin{aligned} 1000 \text{ MB} \\ = 1 \text{ GB} \end{aligned}$$

$$\begin{aligned} &32 + 40 + 48 \\ &= 120 \text{ Megabits} \times \frac{60 \times 60}{1000} \\ &= 54 \text{ GB second} \end{aligned}$$



$$S = \frac{D}{t} \quad t = \frac{D}{S}$$

time = $\frac{\text{distance}}{\text{speed}}$

$$= \frac{9000 \times 1000}{1.5 \times 10^8}$$

$$= \frac{9 \times 10^6}{1.5 \times 10^8} = \frac{6}{10^2} = 6 \times 10^{-2} \text{ sec}$$

$1 \text{ sec} = 1000 \text{ ms}$
 $10^2 = 100$
 $6 \times 10^{-2} \text{ sec} = 60 \text{ milliseconds}$

$$t = \frac{D}{S} = \frac{9 \times 10^6}{3 \times 10^8} = \frac{3}{10^2} = 30 \text{ milliseconds}$$

$$60 + 30 \rightarrow 90 \text{ milliseconds} \times 2$$

$$= 180 \text{ milliseconds}$$

$$\frac{64 \times 2 \times 2}{2^8 \text{ GHz DDR}} = 32$$

$\begin{array}{r l} 16 & 172 \\ \hline 16 & 10 \\ \hline & 0 \end{array}$	remainder 12 C 10 A decimal hexadecimal	$\begin{array}{r l} 16 & 254 \\ \hline 16 & 15 \\ \hline & 0 \end{array}$	14 E 15 F
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AC. 10. FE. 0A

$\begin{array}{r l} 16 & 16 \\ \hline & 1 \end{array}$	remainder 0	$\begin{array}{r l} 16 & 10 \\ \hline & 0 \end{array}$	10 A
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